



University of Colombo School of Computing

SCS 1304 - Problem Solving Strategies and Computation Approaches

Lab Sheet 02

Answer the following questions based on the given data.

- 01.** You are given a dataset containing examination results of university students for one academic year. The dataset includes the following columns: Month, Number of Students, Average Assignment Marks, Average Final Exam Marks, and Overall Pass Percentage. Your task is to identify patterns in the data to help the academic department make informed decisions.

Month	Number of Students	Average Assignment Marks	Average Final Exam Marks	Overall Pass Percentage
January	120	68	70	75%
February	125	70	72	78%
March	130	72	74	80%
April	128	69	71	77%
May	135	74	76	82%
June	140	75	78	85%
July	138	73	75	81%
August	145	76	79	86%
September	150	78	81	88%
October	148	77	80	87%
November	152	79	82	89%
December	155	80	84	91%

- A. Identify the months with the highest and lowest overall pass percentages. What patterns can you observe regarding student performance throughout the year?
- B. Compare the number of students each month. Is there a trend in student participation over the months?

- C. Analyze the consistency of the average assignment marks and final exam marks across the months. What does this indicate about student academic performance?
 - D. Based on the observed patterns, predict the overall pass percentage and average final exam marks for the next month (January of the next year). Explain your reasoning.
 - E. Suggest two strategies the academic department could implement to further improve student performance based on the identified patterns.
 - F. Discuss how recognizing these patterns can help academic decision-making. Provide an example of a decision that could be influenced by these patterns.
2. You are tasked with developing a **new hospital appointment management system**. The system should **allow patients to register, book appointments, cancel appointments, and manage medical records**. To build this system effectively, you need to break down the problem into smaller, manageable parts (functional decomposition).
- A. List the major functions that the **hospital appointment management system** must perform. For each major function identified, break it down further into sub-functions or tasks. Provide at least two sub-functions for each major function
 - B. Explain how functional decomposition helps in managing the complexity of the system development. Provide an example from your decomposition to illustrate this benefit.
3. You are tasked with designing a flowchart for the **“Book Appointment”** function of the hospital appointment management system. The flowchart should clearly illustrate the steps involved from the moment a patient selects a doctor until the appointment is successfully booked and the system is updated.

Hint:

To draw the flowchart for the **“Book Appointment”** function, start by understanding the key steps: patient logs in, searches for available doctors, selects a doctor and time slot, and the system checks appointment availability. If the selected appointment slot is available, the system updates the patient appointment records and doctor schedule, and notifies the patient of successful booking. If the slot is unavailable, the patient is notified about the unavailability and asked to choose another available slot.

4. Create a LeetCode account and try to complete **at least 03** from the following given tasks. Submit screenshots of the successfully completed outputs.

[Two Sum, Palindrome Number, Roman to Integer, Longest Common Prefix, Valid Parentheses.]